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ORIGINAL CONTRIBUTION

Impact of Laterality on Inpatient Outcomes and Complications of Middle Cerebral Artery Acute Ischemic Stroke

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BACKGROUND: Acute ischemic stroke affecting the middle cerebral artery (MCA) remains a major global cause of death and disability. However, it remains unclear whether the laterality of MCA strokes independently influences inpatient outcomes. Despite the fact that the National Institutes of Health Stroke Scale score is heavily weighted toward left-sided acute ischemic stroke, we hypothesize that right MCA strokes (R-MCASs) are associated with worse inpatient outcomes because they would require more severe deficits to amount to a comparable National Institutes of Health Stroke Scale score for a left MCA stroke.

METHODS: A retrospective cross-sectional analysis was conducted using data from the National Inpatient Sample between 2015 and 2022. This nationally representative sample included adult patients (aged ≥18 years) in the United States hospitalized primarily for MCA strokes and with reported National Institutes of Health Stroke Scale scores. Patients were stratified into R-MCAS and left MCA stroke. Outcomes included odds of inpatient mortality, routine discharge, complications, and receipt of reperfusion therapy. Propensity score–based inverse probability of treatment weighting is adjusted for confounding factors. Multivariable logistic regression evaluated adjusted odds ratios (aORs) and 95% CIs for all outcomes.

RESULTS: Among 489 360 acute ischemic stroke hospitalizations, 263 495 (53.8%) were left MCA stroke, and 225 865 (46.2%) were R-MCAS. Following inverse probability of treatment weighting, R-MCASs were significantly more likely to receive endovascular thrombectomy (aOR, 1.26 [95% CI, 1.22–1.30]; $P<0.001$) and less likely to receive intravascular thrombolysis (aOR, 0.965 [95% CI, 0.937–0.994]; $P=0.018$). R-MCAS was associated with significantly increased odds of cerebral edema/herniation (aOR, 1.53 [95% CI, 1.47–1.59]; $P<0.001$), hemorrhagic transformation (aOR, 1.25 [95% CI, 1.20–1.30]; $P<0.001$), and sepsis (aOR, 1.35 [95% CI, 1.25–1.46]; $P<0.001$). Left MCA stroke was associated with a higher likelihood of coma (aOR, 0.920 [95% CI, 0.889–0.952]; $P<0.001$). R-MCAS had significantly higher odds of inpatient mortality (aOR, 1.19 [95% CI, 1.12–1.26]; $P<0.001$) and lower odds of routine discharge (aOR, 0.786 [95% CI, 0.760–0.812]; $P<0.001$).

CONCLUSIONS: Patients with R-MCAS were more likely to receive endovascular thrombectomy and less likely to receive intravascular thrombolysis. R-MCASs were also associated with worse discharge disposition and increased likelihood of inpatient mortality. The National Institutes of Health Stroke Scale bias toward assessment and valuation of patients' language abilities may place patients with R-MCAS at increased risk of adverse poststroke outcomes and complications. Recognition of the subtle signs of nondominant hemisphere strokes may lead to more rapid detection, effective treatment, and improved outcomes.

GRAPHIC ABSTRACT: A graphic abstract is available for this article.

Key Words: brain edema ■ comorbidity ■ ischemic stroke ■ middle cerebral artery ■ thrombectomy

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Nonstandard Abbreviations and Acronyms

AIS	acute ischemic stroke
aOR	adjusted odds ratio
EVT	endovascular thrombectomy
***ICD-10*-CM**	*International Classification of Diseases, Tenth Revision*, Clinical Modification
IVT	intravascular thrombolysis
L-MCAS	left middle cerebral artery stroke
MCA	middle cerebral artery
NIHSS	National Institutes of Health Stroke Scale
NIS	National Inpatient Sample
R-MCAS	right middle cerebral artery stroke

Acute ischemic stroke (AIS) is a major global cause of death and disability, significantly burdening health care systems worldwide, with over 12 million new cases and ≈6.5 million deaths reported annually.[1] However, it remains unclear whether the laterality of middle cerebral artery (MCA) strokes independently influences inpatient outcomes, such as mortality, discharge disposition, reperfusion therapy utilization, and complication rates. Previous studies have shown that right MCA strokes (R-MCASs) were associated with higher odds of acute complications, such as malignant brain edema and cardiovascular events, and longer hospitalizations.[2,3] Worse outcomes in R-MCAS may be explained by differences in patient presentation. Patients with R-MCAS tend to neglect some of their deficits, which may delay presentation. Left MCA strokes (L-MCASs) often present with aphasia, a dramatic and rapidly recognized symptom. Prehospital and clinical recognition of this symptom may drive expedited treatment of L-MCAS, thereby potentially leading to lower acute complications and shorter hospitalizations. In contrast, Ramzan et al[4] found that there were no significant differences in odds of R-MCAS versus L-MCAS reperfusion success, symptomatic intracranial hemorrhagic conversion, functional independence, or short-term mortality after mechanical thrombectomy between the 2 groups.

To investigate these issues, this retrospective study utilizes the National Inpatient Sample (NIS) database to compare inpatient mortality, discharge patterns, reperfusion therapy use, complication rates, and comorbidity profiles between R-MCAS and L-MCAS. We hypothesize that R-MCASs are associated with worse inpatient outcomes, complications, and discharge dispositions due to suboptimal prehospital and clinical stroke recognition and assessment. Addressing the uncertainties and potential biases within clinical stroke assessment could lead to improved inpatient outcomes for individuals who experience AIS.

METHODS

Data Availability Statement

Data, analytic methods, and study materials will be made available to other researchers. This material will be available in a repository managed and maintained by Dr Fawaz Al-Mufti and his research group. To access data, analytic methods, and study materials, please contact the corresponding author.

Data Source and Patient Selection

A retrospective cross-sectional study was completed using the NIS database, the largest publicly available all-payer inpatient care database in the United States. Developed by the Agency for Healthcare Research and Quality as part of the Healthcare Cost and Utilization Project, the NIS approximates a 20% stratified sample of discharges from community hospitals, covering over 97% of the US population and including ≈7 million hospitalizations annually.[5,6] The data are deidentified per the standard defined in Section §164.514 (a) of the HIPAA Privacy Rule, and secondary analysis of existing data does not involve intervention or any interaction with human subjects; therefore, IRB approval was not required for this research study.[7] This study conforms to the Strengthening the Reporting of Observational Studies in Epidemiology reporting guidelines for observational cross-sectional studies.[6,8]

Hospitalizations with a primary diagnosis of AIS between October 1, 2015, and December 31, 2022, were identified among adult patients (aged ≥18 years) using diagnostic codes from the *International Classification of Diseases*, *Tenth Revision*, Clinical Modification (*ICD-10*-CM) guidelines (Table S1).[6,9] R-MCAS and L-MCAS were differentiated based on *ICD-10*-CM codes capturing laterality of the infarct. Reperfusion therapy, specifically intravascular thrombolysis (IVT) and endovascular thrombectomy (EVT), was identified using procedure codes from the *International Classification of Diseases, Tenth Revision*, Procedure Coding System (Table S1), and specific comorbidities and complications were captured using *ICD-10*-CM codes.[6,10] Admissions before October 2015 were excluded due to inconsistency following the transition to *ICD-10*-CM coding in the NIS.

Data Characteristics and Outcomes Measured

According to the *ICD-10*-CM Official Guidelines for Coding and Reporting FY2024, the National Institutes of Health Stroke Scale (NIHSS) codes (R29.7-) can be used in conjunction with acute stroke codes (I60–I63) to identify the patient's neurological status and stroke severity.[6] NIHSS scores were categorized into severity groups: no symptoms (score 0), minor (1–4), moderate (5–15), moderate to severe (16–20), and severe (21–42), in accordance with the Centers for Medicare & Medicaid Services guidelines.[6,11–13] Hospitalizations with no reported NIHSS score were excluded from this analysis.

Primary outcomes measured were the odds of inpatient mortality and routine discharge (defined as home discharge with self-care) in R-MCAS in comparison with L-MCAS hospitalizations. Routine discharge has previously demonstrated a high concordance with a modified Rankin Scale score ≤2 assessed at 90 days, representative of minimal to no disability.[6,14,15] Secondary outcomes included the odds of reperfusion

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therapy, stroke-related complications, presentation, odds of presenting comorbidities, and treatment with hemicraniectomy or tracheostomy.

Statistical Analysis

In accordance with Healthcare Cost and Utilization Project guidelines, all analyses used data weighted to produce national estimates.[15] Descriptive statistics summarized patient demographics and hospital characteristics. Data elements were categorized into subgroups for analysis (Table S2). Median and interquartile ranges for age, length of stay, total charges, and NIHSS scores were calculated to determine population-wide characteristics. Multivariable logistic regression was used to examine the adjusted odds of receiving EVT, IVT, hemicraniectomy, and tracheostomy in the R-MCAS population compared with the L-MCAS population. Propensity score–based inverse probability of treatment weighting is adjusted for confounding factors based on stroke severity, patient demographics, and hospital characteristics between the 2 groups. Propensity scores were estimated using a survey-weighted logistic regression model predicting R-MCAS status from baseline covariates: age, sex, race/ethnicity, median household income quartile, insurance status, Elixhauser comorbidity index, NIHSS score, presence of cerebral edema/herniation, hospital region, urban-rural classification, and EVT or IVT administration. Stabilized weights were calculated to reduce variance inflation, and extreme weights were capped at the 99th percentile to mitigate the influence of outliers.[6]

Weighted multivariable logistic regression models were performed using the inverse probability of treatment weighting–adjusted sample to assess the relationship between infarct laterality and each outcome (routine discharge and inpatient mortality). The models included R-MCAS status as the primary independent variable and adjusted for age, sex, race, Elixhauser comorbidity index, NIHSS score, presence of cerebral edema/herniation, and treatment group to ensure doubly robust estimation. Adjusted odds ratios (aORs) with 95% CIs were reported. Following this, multivariable logistic regression was performed on separate inverse probability of treatment weighting–adjusted samples (excluding cerebral edema/herniation as a variable) to determine the adjusted odds of specified complications and comorbidities, with reported aORs and 95% CIs for each condition. Finally, a sensitivity analysis was conducted by reconducting multivariable logistic regression models with the All Patient Refined Diagnosis-Related Group severity score replacing the NIHSS score in propensity weighting. The sensitivity analysis was conducted to clarify whether laterality itself or NIHSS bias drives any observed differences. Furthermore, to assess the potential effect of selection bias, a second sensitivity analysis was conducted, including all eligible patients regardless of whether an NIHSS score was recorded.[6]

Missing Data

In 5.5% of patient demographic information, there was identified missing data. These data were handled using multiple imputation with 5 iterations as per the recommendations of the Healthcare Cost and Utilization Project for the management of missing data.[6,16]

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RESULTS

Demographics and Comorbidities

A total of 790 190 MCA-AIS hospitalizations were initially reported, with 422 115 L-MCAS (53.4%) and 368 075

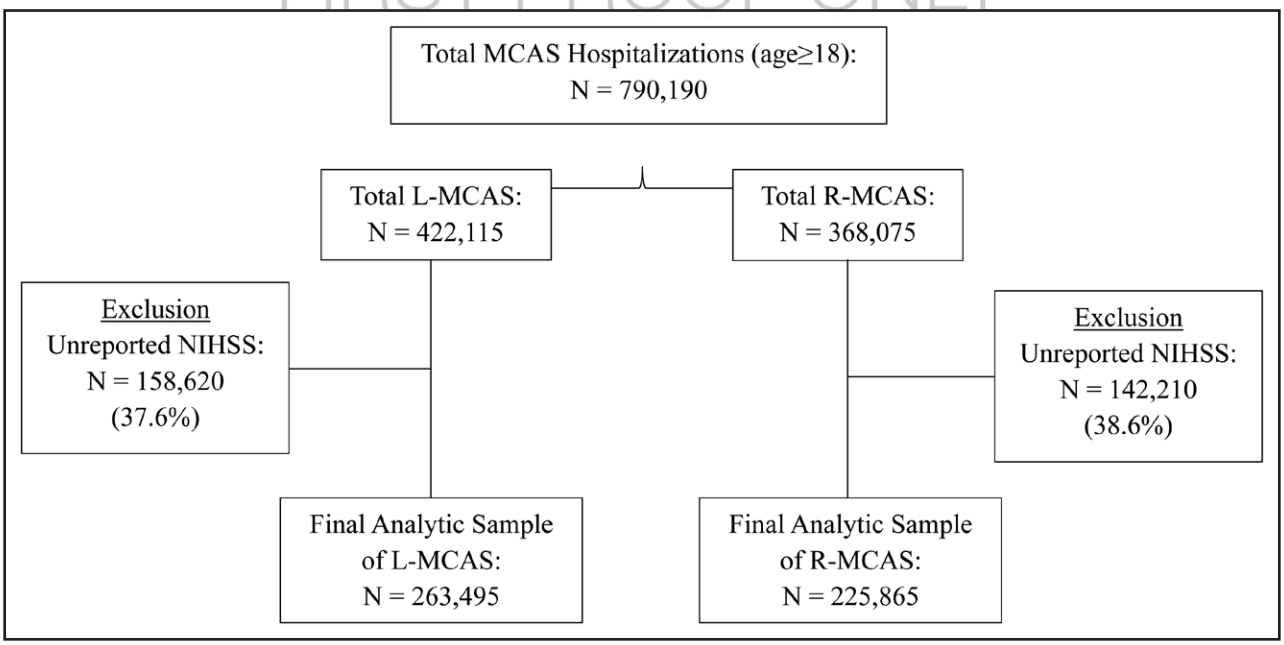

Figure 1. Study population flow diagram: all middle cerebral artery acute ischemic hospitalizations in adults, National Inpatient Sample 2015 to 2022.
The exclusion percent is the percent of right middle cerebral artery strokes (R-MCASs) and left middle cerebral artery strokes (L-MCASs) excluded, respectively. MCAS indicates middle cerebral artery stroke; and NIHSS, National Institutes of Health Stroke Scale.

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Table 1. Patient- and Hospital-Level Demographics of Right and Left MCA Strokes

Variable	Category	Right MCA, %	Left MCA, %	*P* value
No. of patients		225 865	263 495	
Age, y	18–39	6285 (2.8)	7675 (2.9)	0.157
	40–59	43 560 (19.3)	49 660 (18.8)	
	60–79	106 155 (47.0)	125 000 (47.4)	
	>80	69 865 (30.9)	81 160 (30.8)	
Sex	Male	108 315 (48.0)	127 115 (48.2)	0.373
	Female	117 550 (52.0)	136 380 (51.8)	
Race	White	152 885 (67.7)	181 690 (69.0)	<0.001*
	Black	39 210 (17.4)	44 210 (16.8)	
	Hispanic	18 490 (8.2)	20 435 (7.8)	
	Asian or Pacific Islander	7450 (3.3)	7925 (3.0)	
	Native American	905 (0.4)	1200 (0.5)	
	Other	6925 (3.1)	8035 (3.0)	
Income	Quartile 1	67 280 (29.8)	78 190 (29.7)	0.914
	Quartile 2	58 560 (25.9)	68 835 (26.1)	
	Quartile 3	55 275 (24.5)	64 255 (24.4)	
	Quartile 4	44 750 (19.8)	52 215 (19.8)	
Insurance	Medicare	144 955 (64.2)	170 320 (64.6)	0.180
	Medicaid	22 320 (9.9)	24 840 (9.4)	
	Private	43 465 (19.2)	50 950 (19.3)	
	Self-pay	8945 (4.0)	10 075 (3.8)	
	No charge	595 (0.3)	745 (0.3)	
	Other	5585 (2.5)	6565 (2.5)	
NIHSS severity	No stroke symptoms	13 820 (6.1)	14 115 (5.4)	<0.001*
	Minor	59 960 (26.5)	68 605 (26.0)	
	Moderate	95 600 (42.3)	95 760 (36.3)	
	Moderate to severe	31 660 (14.0)	32 930 (12.5)	
	Severe	24 825 (11.0)	52 085 (19.8)	
Hospital location	New England	10 040 (4.4)	12 400 (4.7)	0.297
	Middle Atlantic	29 695 (13.1)	35 035 (13.3)	
	East North Central	35 015 (15.5)	40 815 (15.5)	
	West North Central	16 430 (7.3)	18 950 (7.2)	
	South Atlantic	49 910 (22.1)	57 385 (21.8)	
	East South Central	17 830 (7.9)	21 200 (8.0)	
	West South Central	25 885 (11.5)	30 650 (11.6)	
	Mountain	12 275 (5.4)	14 440 (5.5)	
	Pacific	28 785 (12.7)	32 620 (12.4)	
Teaching status	Rural	6280 (2.8)	7140 (2.7)	0.792
	Urban nonteaching	26 110 (11.6)	30 520 (11.6)	
	Urban teaching	193 475 (85.7)	225 835 (85.7)	
Hospital bed size	Small	26 300 (11.6)	31 125 (11.8)	0.275
	Medium	53 780 (23.8)	63 600 (24.1)	
	Large	145 785 (64.5)	168 770 (64.1)	
Urban-rural classification	Central counties of metro areas of ≥1 million population	65 950 (29.2)	77 610 (29.5)	0.259
	Fringe counties of metro areas of ≥1 million population	53 690 (23.8)	63 225 (24.0)	
	Counties in metro areas of 250 000–999 999 population	50 680 (22.4)	57 655 (21.9)	
	Counties in metro areas of 50 000–249 999 population	19 995 (8.9)	22 805 (8.7)	

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Table 1. Continued

Variable	Category	Right MCA, %	Left MCA, %	*P* value
	Micropolitan counties	20 300 (9.0)	24 125 (9.2)	
	Not metropolitan or micropolitan counties	15 250 (6.8)	18 075 (6.9)	
APR-DRG risk of mortality	Minor likelihood of dying	48 050 (21.3)	51 255 (19.5)	<0.001*
	Moderate likelihood of dying	75 005 (33.2)	90 485 (34.3)	
	Major likelihood of dying	50 860 (22.5)	57 105 (21.7)	
	Extreme likelihood of dying	51 950 (23.0)	64 650 (24.5)	
Treatment group	Neither	135 790 (60.1)	155 855 (59.1)	<0.001*
	IV thrombolysis alone	36 830 (16.3)	47 935 (18.2)	
	EVT without IV thrombolysis	31 675 (14.0)	35 690 (13.5)	
	IV thrombolysis+EVT	21 570 (9.5)	24 015 (9.1)	
Discharge disposition	Routine discharge	62 465 (27.7)	77 045 (29.2)	<0.001*
	Transfer to short-term hospital	6140 (2.7)	6765 (2.6)	
	other transfer	108 570 (48.1)	121 560 (46.1)	
	Home health care	31 875 (14.1)	38 345 (14.6)	
	AMA	1770 (0.8)	1675 (0.6)	
	Inpatient death	15 040 (6.7)	18 090 (6.9)	
	Discharged alive, destination unknown	5 (0.0)	15 (0.0)	
Comorbidity burden	Elixhauser comorbidity index, mean (SD)	4.62 (2.00)	4.89 (2.01)	<0.001*
	Length of stay, d; mean (SD)	6.56 (7.99)	6.39 (7.52)	<0.001*
	Total charges, dollars; mean (SD)	112 792.39 (130 938.03)	108 354.42 (120 709.12)	<0.001*
	Age, y; mean (SD)	70.36 (14.26)	70.40 (14.25)	0.612
	NIHSS score, mean (SD)	9.95 (7.87)	11.17 (8.95)	<0.001*

AMA indicates against medical advice; APR-DRG, All Patient Refined Diagnosis-Related Group; EVT, endovascular thrombectomy; IV, intravascular; MCA, middle cerebral artery; and NIHSS, National Institutes of Health Stroke Scale.

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R-MCAS (46.6%). Of those, 158 620 R-MCAS (37.6%) and 142 210 (38.6%) were removed due to unrecorded NIHSS scores. Included in the final analytical sample, 489 360 MCA-AIS hospitalizations with reported NIHSS scores were identified (Figure 1). Of these, 263 495 (53.8%) were L-MCAS, and 225 865 (46.2%) were R-MCAS. R-MCAS hospitalizations had lower NIHSS scores (9.95±7.87 versus 11.17±8.95; $P<0.001$), lower Elixhauser comorbidity index (4.62±2.00 versus 4.89±2.01; $P<0.001$), higher length of stay (days; 6.56±7.99 versus 6.39±7.52; $P<0.001$), and higher total charges ($112 792.39±130 938.03 versus $108 354.42±120 709.12; $P<0.001$). The average age in both groups was equivalent (years; 70.36±14.26 versus 70.40±14.25; $P=0.612$), and an equivalent number of patients were female (52.0% versus 51.8%; $P=0.373$; Table 1).

R-MCAS hospitalizations presented with higher odds of heart failure (aOR, 1.20 [95% CI, 1.16–1.24]; $P<0.001$), valvular disease (aOR, 1.16 [95% CI, 1.11–1.21]; $P<0.001$), and coronary artery disease (aOR, 1.05 [95% CI, 1.02–1.08]; $P=0.003$). Conversely, R-MCAS hospitalizations were less likely to have hypertension (aOR, 0.914 [95% CI, 0.889–0.940]; $P<0.001$) and dyslipidemia (aOR, 0.935 [95% CI, 0.910–0.960]; $P<0.001$; Table 2; Figure 2).

Treatment Modalities and In-Hospital Complications

R-MCASs were significantly more likely to receive EVT (aOR, 1.26 [95% CI, 1.22–1.30]; $P<0.001$), hemicraniectomy (aOR, 2.84 [95% CI, 2.52–3.21]; $P<0.001$), and tracheostomy (aOR, 1.62 [95% CI, 1.43–1.84]; $P<0.001$) but less likely to receive IVT (aOR, 0.965 [95% CI, 0.937–0.994]; $P=0.018$). R-MCAS also displayed increased odds of nearly all queried complications, including cerebral edema/herniation (aOR, 1.53 [95% CI, 1.47–1.59]; $P<0.001$), hemorrhagic transformation (aOR, 1.25 [95% CI, 1.20–1.30]; $P<0.001$), and sepsis (aOR, 1.35 [95% CI, 1.25–1.46]; $P<0.001$). The only complication with lower likelihood in the R-MCAS group was coma (aOR, 0.920 [95% CI, 0.889–0.952]; $P<0.001$; Table 3; Figure 3).

Inpatient Outcomes

Following inverse probability of treatment weighting adjustment, R-MCAS had significantly higher odds of inpatient mortality (aOR, 1.19 [95% CI, 1.12–1.26]; $P<0.001$) and lower odds of routine discharge (aOR, 0.786 [95% CI, 0.760–0.812]; $P<0.001$; Table 3).

Table 2. Adjusted Odds Ratios of Acute Ischemic Stroke–Related Comorbidities in Right Versus Left MCA Strokes

	Right MCA, %s	Left MCA, %	Odds ratio (95% CI)	*P* value
No. of Patients	225 865	263 495		
Comorbidity				
Anemia	32 695 (14.5)	37 105 (14.1)	1.12 (1.08–1.16)	<0.001*
Arrhythmias	26 535 (11.7)	30 820 (11.7)	1.06 (1.02–1.10)	0.004*
Hypertension	154 900 (68.6)	182 775 (69.4)	0.914 (0.889–0.940)	<0.001*
Heart failure	45 455 (20.1)	52 460 (19.9)	1.20 (1.16–1.24)	<0.001*
Valvular disease	22 010 (9.7)	24 405 (9.3)	1.16 (1.11–1.21)	<0.001*
Dyslipidemia	137 580 (60.9)	163 865 (62.2)	0.935 (0.910–0.960)	<0.001*
Obesity	36 760 (16.3)	42 380 (16.1)	1.09 (1.05–1.13)	<0.001*
Diabetes	75 040 (33.2)	89 100 (33.8)	1.04 (1.01–1.07)	0.015*
Chronic kidney disease	39 845 (17.6)	46 680 (17.7)	1.10 (1.06–1.14)	<0.001*
COPD	23 370 (10.3)	26 700 (10.1)	1.14 (1.09–1.19)	<0.001*
Peripheral vascular disease	7275 (3.2)	8920 (3.4)	1.00 (0.935–1.08)	0.948
Coronary artery disease	51 580 (22.8)	60 510 (23.0)	1.05 (1.02–1.08)	0.003*
Smoking	44 015 (19.5)	49 455 (18.8)	1.04 (1.00–1.07)	0.031*
Long-term anticoagulant use	32 105 (14.2)	37 140 (14.1)	1.04 (1.01–1.08)	0.024*
Long-term antiplatelet use	70 655 (31.3)	81 955 (31.1)	0.980 (0.954–1.01)	0.158
Depression	23 815 (10.5)	27 675 (10.5)	1.08 (1.04–1.13)	<0.001*

COPD indicates chronic obstructive pulmonary disease; and MCA, middle cerebral artery.
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NIHSS Sensitivity Analysis

In the primary sensitivity analysis, R-MCAS retained significantly lower odds of routine discharge and higher odds of most complications. However, the odds of inpatient mortality did not retain a significant difference. In the second sensitivity, including all patients regardless of whether an NIHSS score was recorded, odds ratios of outcomes were similar in magnitude and direction compared with the primary sensitivity analysis, with less than a 3% change in odds ratios for all queried outcomes (Table 4).

DISCUSSION

In our cohort of 489 360 patients, 263 495 (53.8%) had L-MCAS, and 225 865 (46.2%) had R-MCAS. R-MCAS was significantly less likely to result in strokes with severe NIHSS scores. After adjusting for patient- and hospital-level characteristics and NIHSS scores, R-MCAS was associated with a higher likelihood of inpatient mortality, receipt of EVT, hemicraniectomy, and tracheostomy. R-MCAS was associated with a lower likelihood of routine discharge and decreased use of IVT. R-MCAS was also associated with increased risk of most queried complications, including cerebral edema/herniation, hemorrhagic transformation, and sepsis. Following sensitivity analysis, R-MCAS retained significantly decreased odds of routine discharge and increased odds of most complications; however, odds of inpatient mortality became insignificant. There were no significant differences in age and sex in this cohort. The majority of patients with both R-MCAS and L-MCAS were aged 60 to 79 years and female. R-MCAS had increased odds of presentation with existing heart failure, valvular disease, and coronary artery disease but paradoxically lower odds of presentation with hypertension and dyslipidemia.

Mortality and Discharge Disposition

In our cohort, R-MCAS was associated with increased odds of inpatient mortality and discharge disposition. However, in a retrospective cohort study by Li et al,[2] stroke laterality did not have an effect on 1-month mortality or functional outcomes after adjusting for age and stroke severity in a sample size of 314 patients. This earlier study and our cohort both showed that R-MCAS were associated with lower NIHSS scores, as the NIHSS score is weighted toward L-MCAS due to its emphasis on language deficits. While prior literature demonstrated no difference in 10-year outcome survival regardless of laterality, our study shows that laterality may have an effect on inpatient outcomes.[17] These differences may possibly be explained by prior studies being limited to smaller, single-institution samples. Smaller samples may lead to wider CIs and a decreased ability to adequately control for stroke severity.

Reperfusion Therapies

We found that R-MCAS strokes were significantly less likely to be treated with IVT and more likely to be treated

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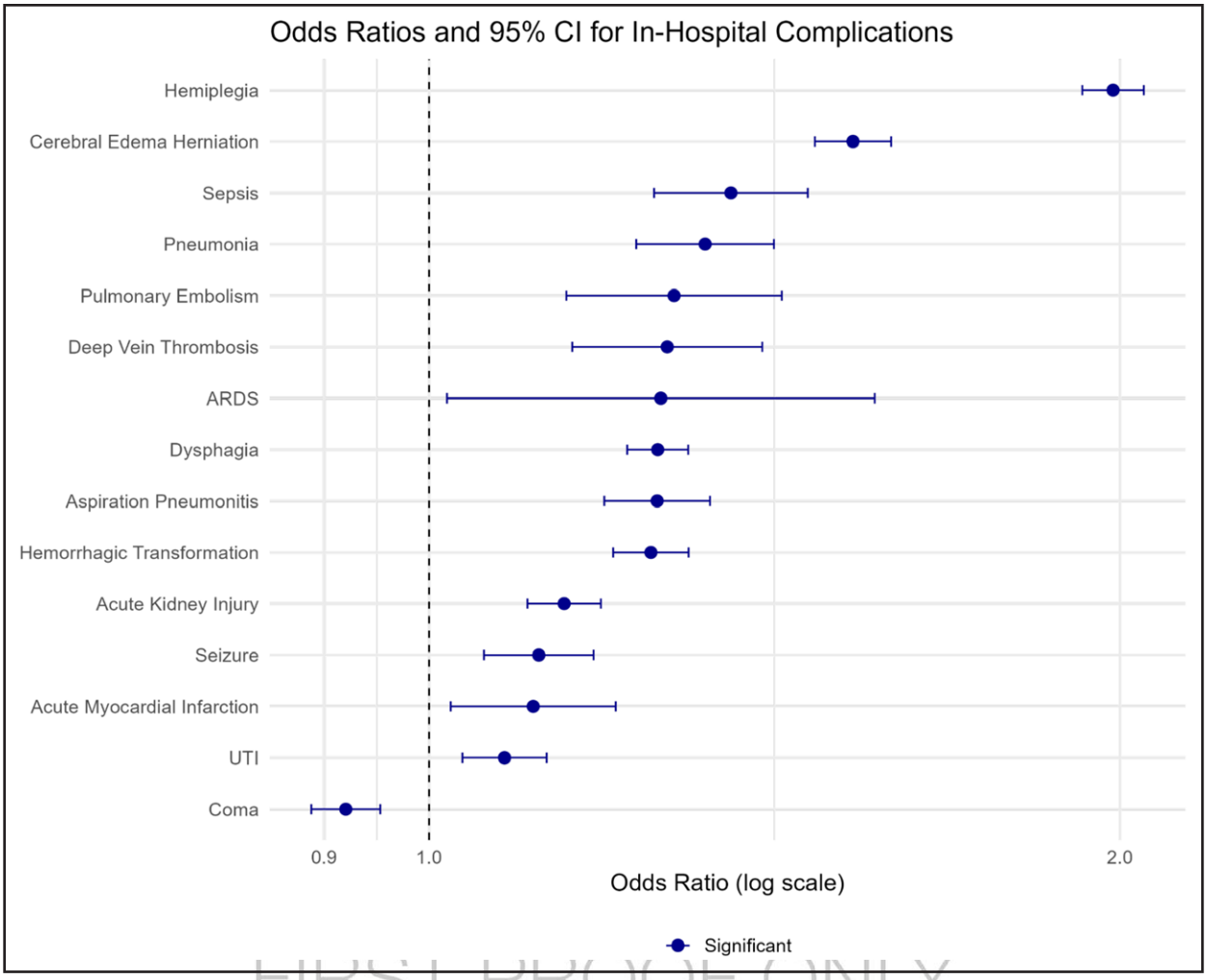

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Figure 2. Forest plot for odds of in-hospital complications in right vs left middle cerebral artery acute ischemic stroke. ARDS indicates Acute respiratory distress syndrome; and UTI, urinary tract infection.

with EVT. However, in a retrospective cohort study with a smaller sample size, hemispheric differences had no effect on either treatment.[2] Another retrospective study found that right hemispheric MCA strokes were less likely to receive tPA and less likely to present at the hospital within 3 hours of symptom onset.[18] Delayed recognition of right hemispheric stroke symptoms by patients and caregivers may factor into the late hospital presentation and decreased likelihood of IVT administration. Language deficits in L-MCAS often present quickly at onset and are among the most easily identified symptoms of ischemic strokes.[19,20] Public-health initiatives focused on promoting earlier recognition of AIS symptoms, such as anosognosia, hemispatial neglect, and prosopagnosia, may lead to more rapid detection, more effective treatment, and improved outcomes of R-MCAS. The NIS database lacks information on stroke onset to arrival time, as well as arrival time to receipt of reperfusion therapy; therefore, future research is warranted to determine whether delays in recognition and treatment are driving differences in reperfusion therapy.

NIHSS Sensitivity Analysis

Following sensitivity analysis, excluding the NIHSS score from propensity weighting, we found that R-MCAS retained decreased odds of routine discharge and most complications but did not retain increased odds of inpatient mortality. These findings may suggest that worse outcomes in R-MCAS may be partially attributed to bias within the NIHSS score and partially attributed to innate differences in stroke morbidity between R-MCAS and L-MCAS. The NIHSS score is utilized to guide implementation of reperfusion therapies, frequency and type of serial stroke imaging, and ICU versus stroke unit patient monitoring.[21–23] The NIHSS score includes 6 of 11 metrics that assess or significantly rely on a patient's language ability.[24] Therefore, dominant hemisphere strokes, which affect the brain's language centers, receive higher average scores than strokes in other brain areas.[25,26] Prior studies have provided evidence for this bias, with L-MCAS strokes receiving higher NIHSS scores compared with equally severe R-MCAS, due to language-related scoring criteria.[27,28] Such biases may result in delayed recognition,

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Table 3. Adjusted Odds Ratios of Right MCA–Related Complications, Outcomes, and Treatments Compared With Left

	Right MCA, %	Left MCA, %	Odds ratio (95% CI)	*P* value
No. of patients	225 865	263 495		
Treatment				
Endovascular thrombectomy	53 245 (23.5)	59 705 (22.6)	1.26 (1.22–1.30)	<0.001*
Intravascular thrombolysis	58 400 (25.8)	71 950 (27.3)	0.965 (0.937–0.994)	0.018*
Hemicraniectomy	3840 (1.7)	2405 (0.9)	2.84 (2.52–3.21)	<0.001*
Tracheostomy	2580 (1.1)	2495 (0.9)	1.62 (1.43–1.84)	<0.001*
Complication				
Sepsis	6105 (2.7)	6515 (2.5)	1.35 (1.25–1.46)	<0.001*
Pneumonia	8065 (3.6)	8585 (3.3)	1.32 (1.23–1.41)	<0.001*
UTI	22 290 (9.9)	25 880 (9.8)	1.08 (1.03–1.13)	<0.001*
Deep vein thrombosis	4280 (1.9)	4370 (1.7)	1.27 (1.15–1.40)	<0.001*
Pulmonary embolism	3025 (1.3)	3295 (1.3)	1.28 (1.15–1.42)	<0.001*
Aspiration pneumonitis	12 815 (5.7)	14 675 (5.6)	1.26 (1.19–1.33)	<0.001*
Dysphagia	54 670 (24.2)	59 115 (22.4)	1.26 (1.22–1.30)	<0.001*
Seizure	12 590 (5.6)	14 515 (5.5)	1.12 (1.06–1.18)	<0.001*
Cerebral edema herniation	34 650 (15.3)	34 635 (13.1)	1.53 (1.47–1.59)	<0.001*
Hemorrhagic transformation	31 685 (14.0)	33 520 (12.7)	1.25 (1.20–1.30)	<0.001*
ARDS	735 (0.3)	815 (0.3)	1.26 (1.02–1.56)	0.034*
Acute kidney injury	32 545 (14.4)	36 805 (14.0)	1.15 (1.10–1.19)	<0.001*
Hemiplegia	156 350 (69.2)	156 250 (59.3)	1.99 (1.93–2.05)	<0.001*
Coma	34 685 (15.4)	44 685 (17.0)	0.920 (0.889–0.952)	<0.001*
Acute myocardial infarction	5205 (2.3)	6105 (2.3)	1.11 (1.02–1.21)	0.013*
Outcome				
Inpatient mortality	62 465 (27.7)	77 045 (29.2)	1.19 (1.12–1.26)	<0.001*
Routine discharge	15 040 (6.7)	18 090 (6.9)	0.786 (0.760–0.812)	<0.001*

ARDS indicates acute respiratory distress syndrome; MCA, middle cerebral artery; and UTI, urinary tract infection.
*XXX.

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less aggressive management, and ultimately poorer outcomes for patients experiencing R-MCAS.[27,28]

To correct this bias, the NIHSS score could be modified to incorporate an expanded assessment of neglect utilizing line bisection tests, clock drawing, or even virtual reality lateralized attention tests.[29,30] Additional assessment of cognitive function associated with the nondominant hemisphere could be included, such as evaluation of emotional prosody, visual and spatial function, and behavioral changes.[19,31] Finally, anosognosia screening could be incorporated into the NIHSS score by evaluating whether patients are aware of any identified impairment. Nevertheless, NIHSS scores do not solely determine the decision to administer thrombolysis, and thorough clinical neurological exams and appropriate imaging are necessary to decide the best treatment strategy for any ischemic stroke. As this is a retrospective study, it is impossible to ascertain whether observed outcome differences are driven directly by scoring bias or if improper scoring leads to suboptimal treatment, which then drives outcome differences. It is possible that both pathways play a role. More specifically, the disparity in NIHSS scores may reflect the difficulty in quantifying right-sided cortical dysfunction as opposed to being a cause of the differences found in management and outcomes between the 2 groups in this study.[28] Further research is warranted to determine whether biased NIHSS scoring leads to suboptimal treatment of R-MCAS, thereby contributing to observed worse outcomes in R-MCAS.

In the second sensitivity analysis, we found that odds ratios of outcomes were similar in magnitude and direction compared with the primary sensitivity analysis, with less than a 3% change in odds ratios for all queried outcomes. These findings suggest that exclusion of patients without NIHSS scores likely did not introduce a meaningful selection bias. Moreover, the similar proportions of patients missing recorded NIHSS scores with R-MCAS and L-MCAS further suggest the likely absence of a relevant selection bias.

Association of R-MCAS With Poststroke Complications

R-MCAS was also associated with increased odds of cerebral edema/herniation, hemorrhagic transformation,

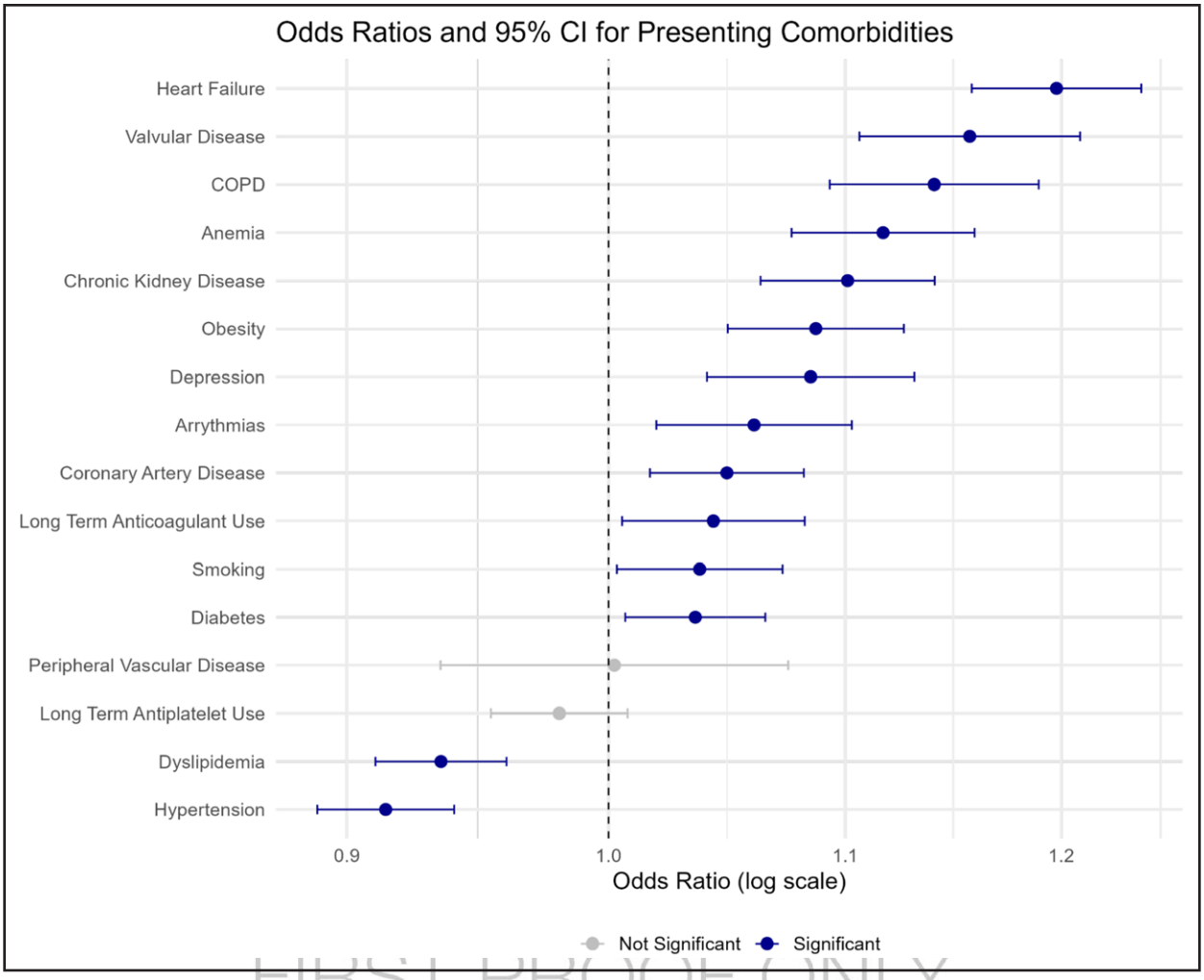

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Figure 3. Forest plot for odds of presenting comorbidities in right vs left middle cerebral artery acute ischemic stroke. COPD indicates chronic obstructive pulmonary disease.

and sepsis. It is impossible to infer causality due to the study's retrospective cross-sectional design. While baseline covariates, including comorbidity burden, were controlled for, unreported or undiagnosed comorbidities may contribute to increased odds of poststroke complications independent of laterality through reverse causality. Differences in clinical monitoring or detection of complications may also contribute to the paradoxical findings; therefore, ascertainment bias should be considered when interpreting these findings. However, these findings, similar to observed differences in in-hospital outcomes, may suggest that R-MCAS have a higher morbidity burden, possibly due to delayed stroke detection and suboptimal in-hospital treatment.

Cardiovascular Comorbidities

In our cohort, patients with R-MCAS had increased odds of presentation with existing heart failure, valvular disease, and coronary artery disease, but they also had paradoxically lower odds of presentation with hypertension and dyslipidemia. These comorbidity patterns may reflect residual confounding from unmeasured vascular risk factor clustering. Alternatively, differential coding practices in administrative data, whereby acute cardiovascular diagnoses are prioritized, while chronic conditions such as hypertension or dyslipidemia are underreported, may contribute to these anomalous findings. Accordingly, these findings should be interpreted with caution, and further studies with more granular clinical data should be conducted to clarify these associations.

Limitations

Several limitations should be taken into consideration when interpreting the findings of this study. First, reliance on *ICD-10*-CM/*International Classification of Diseases, Tenth Revision*, Procedure Coding System codes to identify cases, procedures, and complications may introduce potential coding errors and variability across institutions and between physicians. To mitigate this limitation,

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Table 4. NIHSS Sensitivity Analysis Adjusted Odds Ratios Right MCA Compared With Left (NIHSS Score Control Replaced With APR-DRG Severity Score Control)

	Right MCA, %	Left MCA, %	Odds ratio (95% CI)	*P* value
No. of patients	225 865	263 495		
Treatment				
Endovascular thrombectomy	53 245 (23.5)	59 705 (22.6)	1.09 (1.06–1.13)	<0.001*
Intravascular thrombolysis	58 400 (25.8)	71 950 (27.3)	0.905 (0.905–0.960)	<0.001*
Hemicraniectomy	3840 (1.7)	2405 (0.9)	2.46 (2.37–3.04)	<0.001*
Tracheostomy	2580 (1.1)	2495 (0.9)	1.39(1.21–1.67)	<0.001*
Complication				
Sepsis	6105 (2.7)	6515 (2.5)	1.16 (1.06–1.25)	<0.001*
Pneumonia	8065 (3.6)	8585 (3.3)	1.16 (1.08–1.24)	<0.001*
UTI	22 290 (9.9)	25 880 (9.8)	1.03 (0.985–1.07)	0.214
Deep vein thrombosis	4280 (1.9)	4370 (1.7)	1.20 (1.09–1.32)	<0.001*
Pulmonary embolism	3025 (1.3)	3295 (1.3)	1.17 (1.05–1.30)	<0.001*
Aspiration pneumonitis	12 815 (5.7)	14 675 (5.6)	1.07 (1.03–1.12)	0.001*
Dysphagia	54 670 (24.2)	59 115 (22.4)	1.16 (1.12–1.19)	<0.001*
Seizure	12 590 (5.6)	14 515 (5.5)	1.05 (0.996–1.11)	0.071
Cerebral edema herniation	34 650 (15.3)	34 635 (13.1)	1.53 (1.47–1.59)	<0.001*
Hemorrhagic transformation	31 685 (14.0)	33 520 (12.7)	1.17 (1.12–1.22)	<0.001*
ARDS	735 (0.3)	815 (0.3)	1.09 (0.880–1.35)	0.427
Acute kidney injury	32 545 (14.4)	36 805 (14.0)	1.12 (1.08–1.16)	<0.001*
Hemiplegia	156 350 (69.2)	156 250 (59.3)	1.90 (1.84–1.96)	<0.001*
Coma	34 685 (15.4)	44 685 (17.0)	0.876 (0.845–0.907)	<0.001*
Acute myocardial infarction	5205 (2.3)	6105 (2.3)	1.02 (0.940–1.11)	0.653
Outcome				
Inpatient mortality	62 465 (27.7)	77 045 (29.2)	0.957 (0.904–1.01)	0.128
Routine discharge	15 040 (6.7)	18 090 (6.9)	0.786 (0.760–0.812)	<0.001*

APR-DRG indicates All Patient Refined Diagnosis-Related Group; ARDS, acute respiratory distress syndrome; MCA, middle cerebral artery; NIHSS, National Institutes of Health Stroke Scale; and UTI, urinary tract infection.

*XXX.

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authors utilized the 2023 Healthcare Cost and Utilization Project CCSR files to fulfill desired research queries.

Second, the NIS is limited to information captured during the index hospital admission. Postdischarge events, including long-term mortality and functional recovery, cannot be directly assessed. While our analysis provides insights into the odds of inpatient outcomes in R-MCAS versus L-MCAS, these outcomes do not fully capture the long-term trajectory of patients with stroke. Prior studies have demonstrated that postdischarge complications, rehabilitation access, and delayed functional recovery significantly influence 90-day and long-term outcomes.[32,33] Consequently, further prospective studies and longitudinal retrospective studies, incorporating follow-up data, are warranted to evaluate the impact of stroke laterality on long-term morbidity and functional status.

Next, the NIS database does not report the modified Rankin Scale score at discharge or at 90 days. To account for this limitation, we utilized routine discharge as a surrogate for functional independence with minimal to no disability. Routine discharge is a practical and widely used surrogate for functional independence in administrative data sets.[14,15] However, the likelihood of routine discharge may be influenced by differences in institutional discharge practices, availability of post-acute care facilities, and patient support systems at home. These differences may introduce variability in how well routine discharge reflects true functional independence.

In addition, the NIS database lacks data indicating stroke onset to hospital arrival time, as well as hospital arrival time to receipt of reperfusion therapy. These limitations hinder our ability to substantiate the possibility posited in the discussion that delays in presentation and treatment of R-MCAS may be contributing to lower use of IVT in R-MCAS, despite similar age and comorbidity profiles. Further research is warranted to determine whether R-MCAS and L-MCAS have differences in stroke onset to hospital arrival time and hospital arrival time to reperfusion.

Next, the NIS database does not have sufficient granularity to isolate and control for individual

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nonlanguage components of the NIHSS score; therefore, we were unable to perform a sensitivity analysis, only excluding language-specific components of the NIHSS score. To address this limitation, while still conducting a sensitivity analysis of NIHSS score propensity weighting, we replaced NIHSS score with the All Patient Refined Diagnosis-Related Group severity index in our sensitivity analysis. The All Patient Refined Diagnosis-Related Group severity index is derived from demographics and comorbidity. Therefore, it does not function as an ideal control for stroke severity. However, it allows a sensitivity analysis, excluding any effect of language weighting in the NIHSS score, to be conducted with some control for patient illness at presentation still retained. Also, while patients without reported NIHSS scores were removed from the final analytical sample, the percentages of patients excluded per hemisphere were comparable and likely do not present any significant bias to the analyses conducted. However, to assess the impact of excluding patients without reported NIHSS scores, we performed a second sensitivity analysis that showed consistent effect sizes and directions, with less than a 3% change in odds ratios across outcomes.

Finally, residual confounding after control for comorbidities, as well as differential hospital *ICD-10* coding practices, may have contributed to increased odds of R-MCAS presentation with existing heart failure, valvular disease, and coronary artery disease but paradoxically lower odds of presentation with hypertension and dyslipidemia. It is possible that such artifacts could exaggerate baseline differences between groups and complicate the interpretation of outcome comparisons. Despite these limitations, the use of nationwide data and the large sample size of this study enhance the generalizability and reliability of our findings. Leveraging a comprehensive national database mitigates the potential effects of hospital-specific biases and provides valuable insights into the clinical outcomes of R-MCAS versus L-MCAS.

Conclusions

This retrospective cross-sectional study found that patients with R-MCAS were more likely to experience inpatient mortality and undergo EVT, and less likely to experience routine discharge, as well as receive IVT compared with L-MCAS. R-MCAS was associated with increased odds of major complications, including cerebral edema/herniation, hemorrhage, and sepsis. Last, patients with R-MCAS had lower NIHSS scores, suggesting a bias toward left-sided strokes. Further research should focus on further optimizing the NIHSS score to include right-sided stroke symptoms, such as hemispatial neglect and anosognosia. Public-health efforts to recognize the subtle signs of nondominant hemisphere strokes may lead to more rapid detection and improved outcomes of R-MCAS.

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Supplemental Material

Tables S1–S2
STROBE Statement Checklist

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